MILITARY SPECIFICATION

GRAPHITE, NATURAL, POWDERED, TECHNICAL

This specification has been approved by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1 1 Scope. This specification covers the requirements for the procurement of one type of ground Ceylon-type graphite

2 APPLICABLE DOCUMENTS

2 1 Issues of documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

Military

WS 4503B	Rocket Motor, Mk 56 Mods 0, 1 & 2

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

PUBLICATIONS

Naval Sea Systems Command (Code Ident 10001)

OD 18876	Particle Size Distribution by Sieve Analysis, Determination of

(Copies of standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2 2 Other publications The following documents form a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6810

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American Society for Testing and Materials

ASTM C 561	Standard Method of Test for Ash in Graphite
ASTM C 562	Moisture in Graphite

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa 19103)

CODE OF FEDERAL REGULATIONS

49 CFR 100-199	Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, U S Government Printing Office, Washington, DC 20402 Orders should specify "49 CFR 100-199 (latest revision)")

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn Tariff Order Section, 1616 P Street, Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106 222 South Riverside Plaza, Chicago IL 60606)

(Technical society and technical association specifications and standards are generally available for reference from libraries They are also distributed among technical groups and using Federal agencies)

3 REQUIREMENTS

3.1 General material requirements. The material supplied under this specification shall conform to the following requirements The contractor shall make no changes in processing techniques or other factors affecting the quality of the product without prior approval of the procuring activity

3.1.1 Mark 56 Mod 2 Dual Thrust Rocket Motor procurement When the graphite is procured for the MK 56 Mod 2 Dual Thrust Rocket Motor the requirements for change of material material vendors, processes or equipment shall be those in WS 4503 paragraph 3.1.3 (6.2.1)

3.2 Form. The material shall be a dry powder, free from any indications of caking or lumping.

3.3 Composition The material shall be Ceylon-type natural occurring graphite consisting of a single element carbon

3.4 Color. The material shall be grey-black in color

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3 5 Chemical and mechanical properties. The material shall conform to the following properties

		Min	Max
a	Ash, wt%	--	5
b	Carbon, wt%	95	—
c	Particle size distribution by sieve analysis, wt%		
	-100 mesh 149µm	99	100
	-200 mesh 74µm	65	90
	-325 mesh 44µm	35	60

3.6 Workmanship The material shall be free from all impurities and other defects that could adversely affect its performance. The material shall be uniform in quality and shall be manufactured in accordance with standard manufacturing procedures of the industry.

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements

4.2 Inspection conditions Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions

a Temperature Room ambient 18 to 35°C (65° to 95°F)
b Altitude Normal ground
c Vibration None
d Humidity Room ambient to 95 percent relative, maximum

4 3 Certificate of compliance When specified in the contract (see 6 2), the contractor shall furnish a certificate of compliance stating that the graphite is in conformance with this specification

4 4 Quality conformance inspection

4.4 1 Sampling Four containers shall be randomly selected from each lot and shall be individually sampled. If the lot consists of less than four containers, each of the containers shall be sampled Each sample shall be placed in a separate sampling jar.

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4 4.2 Lot size A lot is defined as a uniform blend of one or more batches, or parts thereof, offered for acceptance at one time.

4 4 3 Tests. Tests and examinations comprising quality conformance inspection are specified in TABLE I Each sample selected per 4.4 1 shall be subjected to the tests and examinations of TABLE I

4 4 3 1 Ash content The ash content shall be determined by the method of ASTM C 561 using material dried in accordance with ASTM C 562.

4.4 3 2 Carbon content The carbon content shall be calculated by subtracting the ash content from the dried sample weight.

4.4 3 3 Sieve analysis The particle size distribution of the material by sieve analysis shall be determined in accordance with OD 18876, except the sample size shall be 25 grams (g) weighed to the nearest 0.01 g, the sieve series stack order shall be 100 200, 325 and the sieve series stack shall be shaken for 45 minutes

4 4 4 Rejection Criteria If any of the individual samples, selected in accordance with 4 4 1 do not conform to the requirements specified herein, or the material is otherwise contaminated, the lot shall be rejected

TABLE I

QUALITY CONFORMANCE INSPECTION

Test	Requirement	Test Method
Packaging and marking	Section 5	Visual
Form	3 2	Visual
Color	3 4	Visual
Ash	3 5 a	4 4.3 1
Carbon content	3 5 b	4 4 3 2
Particle size sieve analysis	3 5 c	4 4 3.3
Workmanship	3 6	Visual

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5. PACKAGING

5 1 Packaging and packing Unless otherwise specified in the contract (see 6 2), packaging and packing shall be level C

5 1.1 Level C Unless otherwise specified in the contract (see 6 2), packaging and packing of the graphite shall be in accordance with standard commercial practice applicable to the type of material The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination Container size shall be as specified in the contract (see 6.2) Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation.

5 2 Marking In addition to any special marking required by the contract (see 6 2), each container shall be marked in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 171-179. Marking shall include, but not be limited to, the following information

a Title, number and date of this specification.

b Manufacturer's name and location

c Material trade name

d Net weight and volume

e Lot number, batch number(s), and date of manufacture.

f Storage conditions

g Toxicity precautions, if applicable.

h Contract number.

6 NOTES

6 1 Intended use The graphite is intended to be used as a propellant ingredient for the Mark 56 Mod 2 Dual Thrust Rocket Motor or other rocket motor applications.

6 2 Ordering data Procurement documents should specify the following

6 2 1 Procurement requirements

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a. Title, number and date of this specification.

b Quantity required.

c Place of delivery

d Inspection conditions when other than as specified (see 4.2)

e Packaging requirements if other than as specified (see 5 1)

f Size of container required (see 5 1.1)

g Any special marking required (see 5.2)

h Changes to material, material vendors, processes and equipment (see 3 1.1)

6 2.2 Contract data requirements Items of deliverable data required by this specification are cited in the following paragraph herein

Paragraph	Data Requirement	Applicable DID*
4.3	Certificate of compliance	DI-E-2121

*DIDs (Data Item Description/DD Form 1664) for the above requirements are documented in the applicable ADL (Authorized Data List) Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6 3 Rounding-off specified limits The following applies to all specified limits in this specification For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29, Standard Recommended Practice for Indicating Which Places of Figures are to be Considered Significant in Specified Limiting Values

6 4 Toxicity and safety The material is considered by CAL-OSHA as a nuisance dust with a threshold limit of 15 million parts per cubic foot Avoid inhalation of the material

6 5 Environmental stability The material shall be able to withstand storage in well sealed containers in a clean dry location at ambient temperature The shelf life of the material shall be at least five years when stored in original closed containers

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6.6 Suggested source of supply. Graphite which has been found satisfactory for use in solid propellant rocket motor applications is material designated Dixon 635 graphite, manufactured by Joseph Dixon Crucible Company, 167 Wayne St., Jersey City, NJ 07303 (FSCM 72657). This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

Custodian
NAVY - OS

Preparing Activity.
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Project Number
6810-NA46

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes" in what way?)*

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